

EDUCATION

- **University of Glasgow** Glasgow, Scotland, UK
M.Sc. Robotics and Artificial Intelligence Sept. 2021 – Sept. 2022
- **University of Mumbai** Mumbai, India
B.Tech. Electronics Engineering June 2015 – June 2019

EXPERIENCE

- **Auriga AI Ltd.** London, UK
Autonomy Engineer (Perception) Jul. 2025 – Present
 - **Perception Systems:** Architected a real-time 3D multi-object tracking pipeline for autonomous vehicles, integrating lane detection and euclidean clustering; reduced end-to-end inference latency by 10–20%.
 - **System Diagnostics:** Designed a robust diagnostics platform with failure detection, sensor health validation, and temporal consistency checks for LiDAR and camera-based perception.
 - **CI/CD & Deployment:** Designed a CI/CD pipeline for autonomy and perception modules using GitHub Actions and CMake for ROS2-based systems.
- **Aidrivers Ltd.** London, UK
Autonomy Engineer Dec. 2024 – May 2025
 - **Autonomy Platform Support:** Contributed to the development and integration of autonomy stack components for an on-board prime mover autonomous vehicle platform.
 - **Diagnostics:** Developed diagnostics for on-board sensors including cameras, LiDARs, and IMUs, monitoring data availability and runtime behaviour.
- **eCom Learning Solutions** Glasgow, Scotland, UK
Software Engineer – Robotics Systems Nov. 2022 – Nov. 2024
 - **Robotic Teleoperation:** Developed an Innovate UK funded VR-based teleoperation system for robotic manipulation (Franka Emika) with real-time control loops.
 - **Digital Twin & Control:** Built digital twin simulations using Azure Digital Twins and Three.js; implemented low-latency telemetry via Azure IoT Hub and SignalR.
- **FLOX** Pune, India
Software Developer – Simulation & Perception Jan. 2021 – June 2021
 - **Robotic Simulation:** Developed Ignition Gazebo system plugins to model drone flight dynamics and sensor-environment interactions.
 - **Perception:** Integrated YOLO-based thermal object detection pipelines for UAVs for autonomous target detection in low-visibility conditions.

OPEN SOURCE & PROJECTS

- **RentShield Pro** Open Source Contributor
github.com/addy1997/rentshield_pro Jan. 2026
 - **Progressive Web App:** Developed during the **Granola DeepMind Hackathon** using **Gemini Multimodal capabilities** to automate property inspections and damage detection via vision-language modeling.
- **Pymujoco** Open Source Contributor
Python bindings for the MuJoCo physics engine 2021 – Present
 - **Simulation:** Contributed to Python-based wrappers for MuJoCo (DeepMind's physics engine), ensuring API parity and enhancing accessibility for RL research.

PUBLICATIONS

- **Journal Article** Jan. 2021
HMM-based phoneme speech recognition system for control of industrial robots
[Link](#)
- **Conference Paper** Mar. 2023
Super-immersive Remote Working via Virtual Reality Controlled Robotics
[Link](#)
- **Master's Thesis** Aug. 2022
Novel View Synthesis from Blurry Images
[Link](#)

ACHIEVEMENTS

- **Granola DeepMind Hackathon (2026):** Selected as one of **80 participants from 600+ applicants** to build AI applications using Gemini Multimodal models at Granola's London headquarters.
- **Mobile World Congress (MWC) 2024:** Developed and exhibited an augmented reality (AR) application at the global showcase in Barcelona, Spain.
- **Scholarship:** Recipient of the Bertelsmann Technology Scholarship by Udacity (2021).

TECHNICAL SKILLS

Languages: Python, C++, C#, JavaScript, TypeScript, $\text{\LaTeX}$
Robotics & Sim: ROS/ROS2, **MuJoCo**, Gazebo, Sim-to-Real, Sensor Fusion, Perception Pipelines
ML & Computer Vision: Gemini Multimodal, PyTorch, TensorFlow, OpenCV, NumPy, SciPy
Cloud & DevOps: Git, GitHub Actions, CI/CD, CMake, Azure Digital Twins
Visualization: Three.js, A-Frame, WebXR, Augmented Reality (AR)